

Course Name: Computer Networks and Internet Protocol

Assignment 1 - Week 1 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1

Map the devices with their associated layer in the TCP/IP model.

- | | |
|-----------|--------------------|
| A. NIC | 1. Physical Layer |
| B. Router | 2. Data Link Layer |
| C. Bridge | 3. Network Layer |
| D. Hub | 4. Transport Layer |

- a) A-2, B-3, C-2, and D-1
- b) A-3, B-3, C-2, and D-1
- c) A-3, B-3, C-1, and D-2
- d) A-2, B-3, C-2, and D-2

Correct Answer: (a)

Detailed solution:

Hub operates at Layer 1 of the TCP/IP model, while Bridge, NIC, and Switch(L2) work with MAC addresses at Layer 2. A Router works with IP addresses and it operates in a network layer.

QUESTION 2

Which of the following is/are true with respect to Bridge?

- (i) Also called “smarter hub”
 - (ii) It filters network traffic based on MAC address
-
- a) Only (i)
 - b) Only (ii)
 - c) Both (i) and (ii)
 - d) Either (i) or (ii)

Correct Answer: (c)

Detailed Solution: A bridge is often called a smarter hub because it forwards and filters network traffic intelligently using MAC addresses, making both statements true.



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QUESTION 3

Which of the following statement(s) are concerned with encapsulation in computer networks?

- I. It involves an addition of a header and trailer to the actual data as it moves down in the TCP/IP protocol stack.
 - II. At the transport layer, encapsulation includes port numbers and at the network layer, encapsulation includes IP addresses.
- a) Only (I)
 - b) Only (II)
 - c) Both (I) and (II)
 - d) Neither (I) nor (II)

Correct Answer: (c)

Detailed solution: Encapsulation is the process of addition of a header and trailer while moving down from upper layer to lower layer like adding source and destination port numbers to the application layer data, adding source and destination IP addresses to the transport layer segment etc.

QUESTION 4

Which of the following is NOT a disadvantage of the circuit switching technique over the packet switching technique in computer networks?

- a) Delay in the establishment of a dedicated connection between two hosts
- b) Inefficient utilization of the resources
- c) Less scalable
- d) Provides a dedicated communication path between two end hosts

Correct Answer: (d)

Detailed solution: Circuit switching techniques establish a dedicated path for communication (which is an advantage), although it remains unutilised in case of no data transmission. It also suffers from greater setup time delay, scalability issues, lack of fault tolerance, etc.

QUESTION 5

What is socket address?

- a) Combination of port and IP address
- b) Combination of MAC and IP address
- c) IP address only

d) None of this

Correct Answer: (a)

Detailed Solution: socket address is the combination of port and IP address. Number socket is an internal terminal for transferring and receiving data/packets at a single node in a computer network.

QUESTION 6

Which layer collects a stream of bits into a frame?

- a) Physical Layer
- b) Network Layer
- c) Data Link Layer
- d) Session Layer

Correct Answer: (c)

Detailed Solution: The Data Link Layer (Layer 2) of the OSI model is responsible for collecting a stream of bits from the physical layer and assembling them into a meaningful unit called a frame.

QUESTION 7

Which transport layer protocol will be chosen by an application that does not require any reliability?

- a) TCP
- b) UDP

Correct Answer: (b)

Detailed solution: User Datagram Protocol (UDP) is a connectionless-oriented protocol and provides no reliability.

QUESTION 8

What is the First search-engine?

- a) Chrome
- b) Internet Explorer
- c) HotBot
- d) Archie

Correct Answer: (d)

QUESTION 9

Choose the statement(s) that correctly describe(s) the roles and responsibilities of the Network Interface Card (NIC) in the computer networks.

- I. NIC takes over the responsibility of segmenting large chunks of data into smaller packets that can be transmitted over the networks.
 - II. It converts digital data from the computer's processor to an analog signal.
- a) Only (I)
 - b) Only (II)
 - c) Both (I) and (II)
 - d) Neither (I) nor (II)

Correct Answer: (c)

Detailed solution: The NIC's primary role at the Data Link Layer is to ensure reliable frame-based transmission of data over a local network, including responsibilities for framing, MAC addressing, error detection, flow control, and managing access to the transmission medium. In computer networking, large send offload (LSO) is a technique for increasing egress throughput of high-bandwidth network connections by reducing CPU overhead. It works by passing a multipacket buffer to the network interface card (NIC). The NIC then splits this buffer into separate packets. The technique is also called TCP segmentation offload (TSO) or generic segmentation offload (GSO) when applied to TCP.

QUESTION 10

What is the size of IP address (IPv4) in bytes?

- a) 4
- b) 32
- c) 16
- d) 10

Correct Answer: (a)

Detailed Solution: IP (IPV4) address is 32 bits long that is 4 bytes.